Lab 6 — Capstone: Smart Site Monitor (Smart Environment Station)

Time: ~18 minutes in the Telecom & Data-Centre Edition, as the alternative to Lab 7 (~45 minutes on your own) **You'll learn:** nothing new, on purpose. This lab is about combining everything from today into one working project — which is what building real IoT systems actually feels like.

> **Telecom & Data-Centre Edition**
>
> **The story:** This is the **Smart Site Monitor** with a screen: it reads the shelter's air, shows the status on the shelter's display, switches extra cooling when it gets hot, and reports `normal` or `ALARM` to the NOC.
>
> **In the workshop:** The default capstone in Module 8 is Lab 7, the Telecom Site Monitor, which adds the battery, the fuel tank, the door and the generator. Do this lab instead if your team prefers the screen version.
>
> **Your team:** Teams of three. Rotate the roles from the last lab: **builder** (keyboard), **analyst** (reads the guide, checks the output), **decision owner** (answers the manager task).
>
> **Extension (fast teams):** Add a door-contact button and make an open door a second reason for `ALARM`.
>
> **Business-case canvas:** After the lab, fill in boxes 6 (costs) and 7 (payback), then prepare your 2-minute pitch.

What you're building

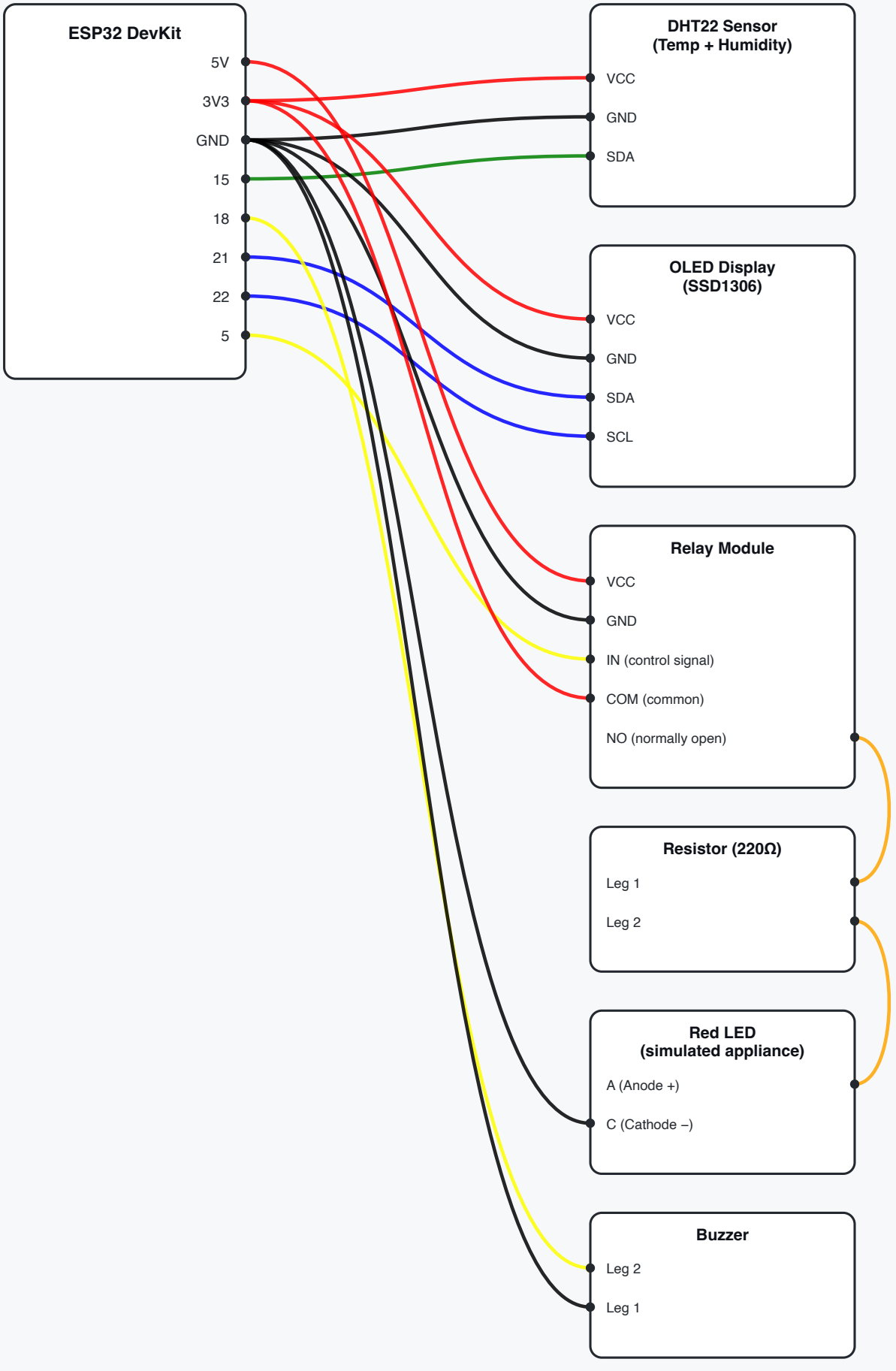
One project that: 1. Reads temperature and humidity from a DHT22 (Lab 2). 2. Shows the live reading and a status message on a small OLED screen (new — but it's just `print()`, aimed at a screen instead of the Serial Monitor). 3. Decides, locally on the device, whether to switch on a relay and buzzer past a threshold (Lab 5 — edge computing). 4. Publishes the reading and status to the cloud over MQTT, so it's also visible remotely (Lab 3).

This mirrors a real product: a local display for whoever is standing next to the device, an automatic local safety reaction that works even if the internet drops, and a cloud feed for anyone monitoring it remotely.

Wiring diagram

Lab 6 — Capstone: Smart Environment Station Wiring

Generated from this lab's real diagram.json — colors match the wires you'll paste into Wokwi



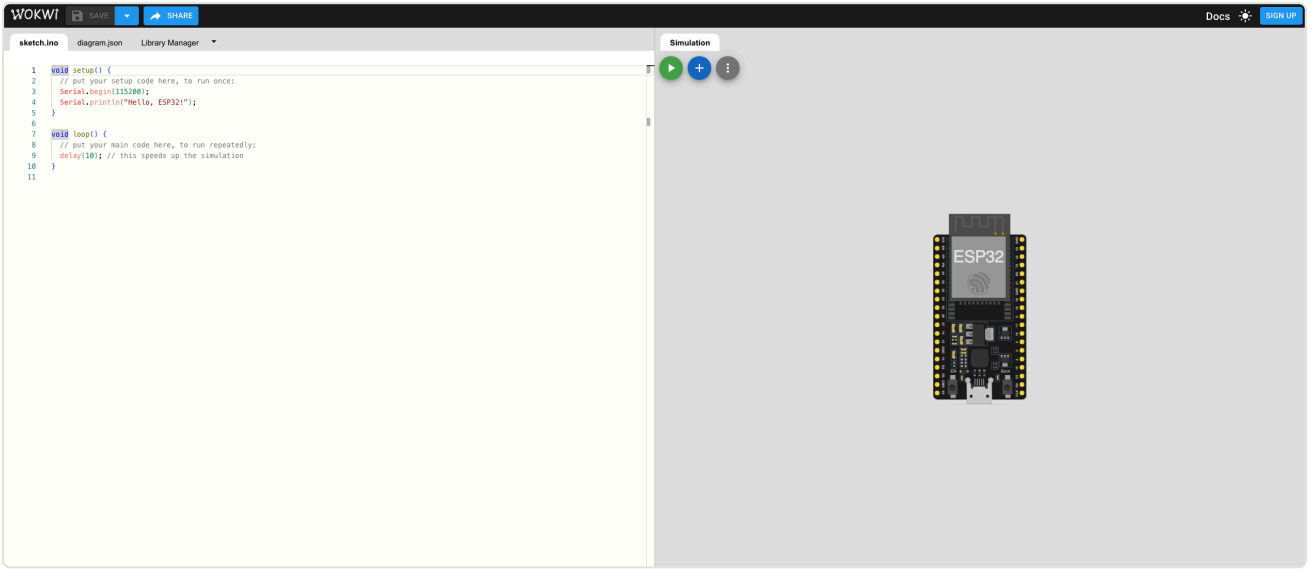
Wire colors match the real diagram.json for this lab — red = power, black = ground, other colors = signal/data.

ESP32 Pin	Wire Color	Connects To
`5V`	red	Relay Module (VCC)
`3V3`	red	DHT22 (VCC), OLED (VCC), Relay Module (COM)
`GND`	black	DHT22 (GND), OLED (GND), Relay Module (GND), simulated-appliance LED (Cathode), Buzzer
`15`	green	DHT22 (data pin)
`21`	blue	OLED (SDA)
`22`	blue	OLED (SCL)
`5`	yellow	Relay Module (IN)
`18`	yellow	Buzzer

Every one of these connections is something you already wired in an earlier lab — the OLED (pins 21/22, a two-wire “I2C” display link) is the only new part today.

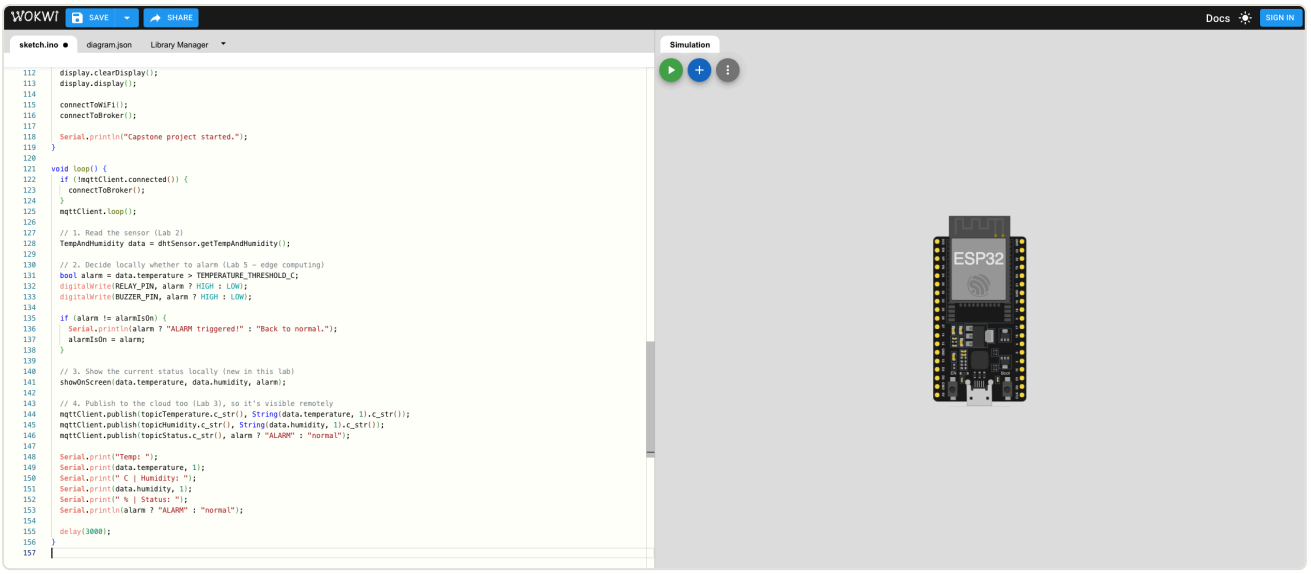
Step-by-step

1. Open a new ESP32 project



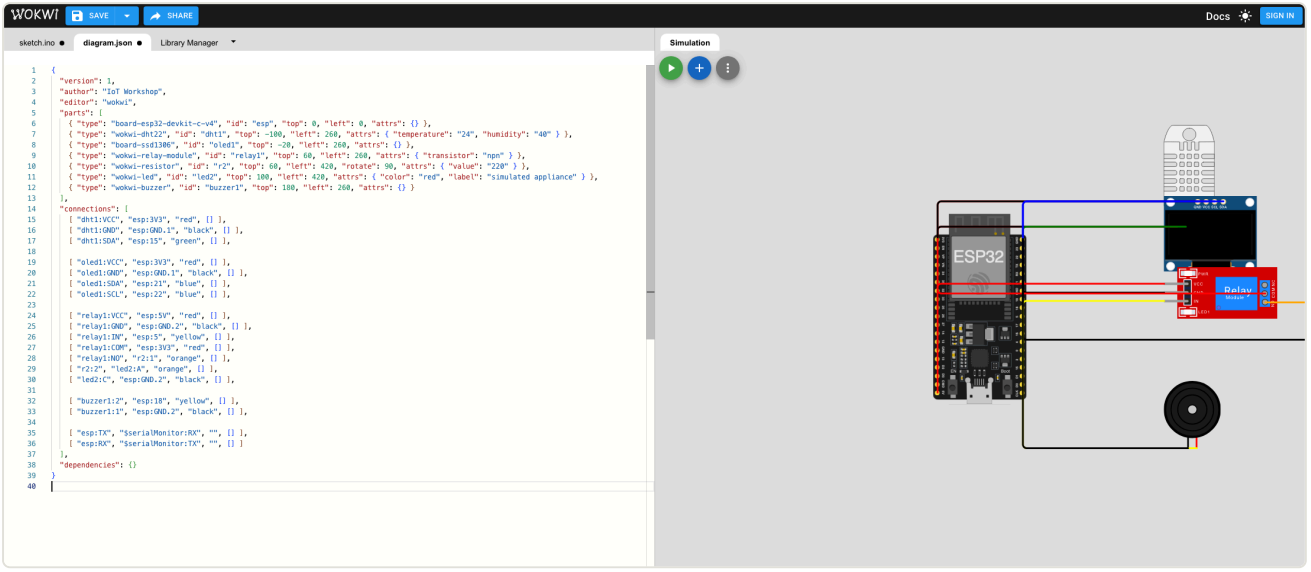
2. Paste the code — and change one line

Copy `sketch.ino` into Wokwi’s **sketch.ino** tab. Before pressing Play, change `YOUR_NAME` near the top to something unique, exactly like in Lab 3.



3. Paste the circuit

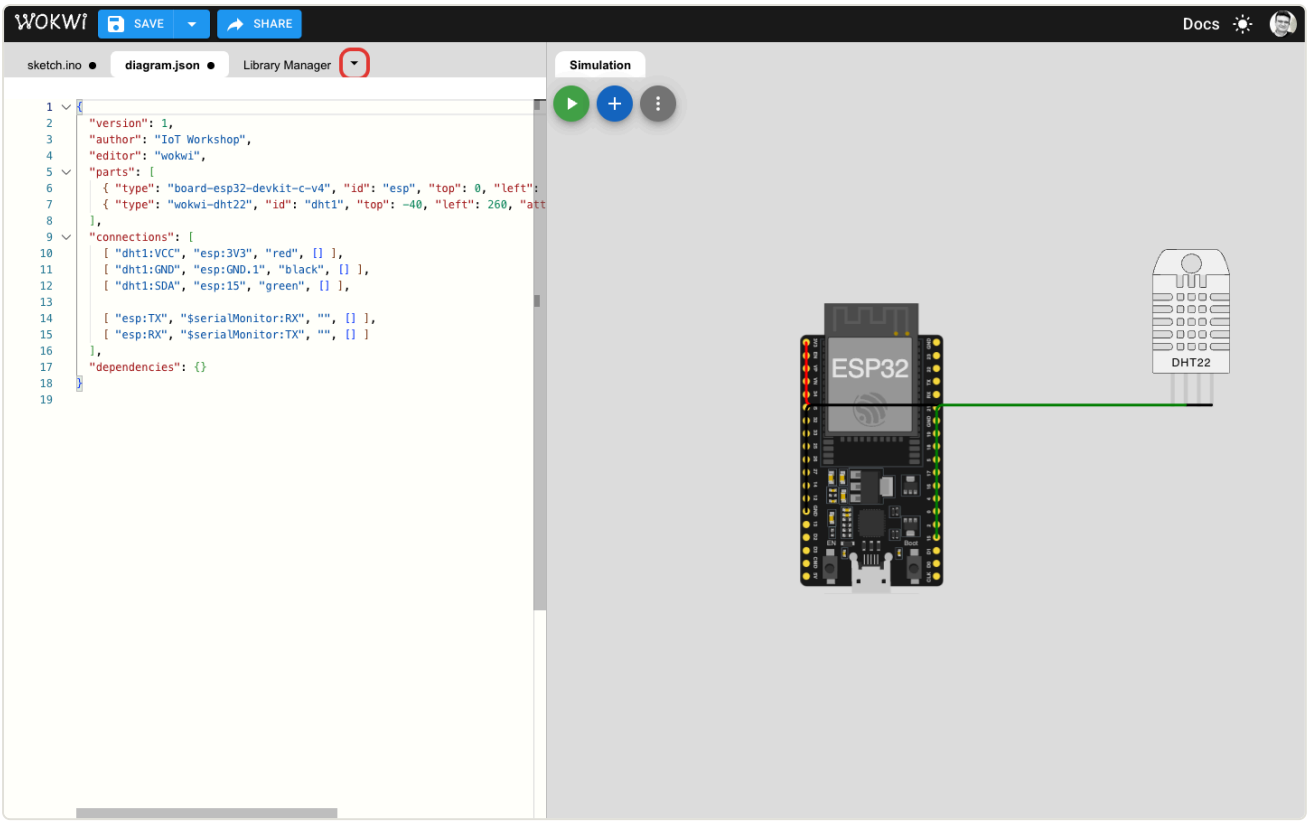
Copy `diagram.json` into Wokwi's **diagram.json** tab. You should see the DHT22, OLED screen, relay module, buzzer, and the "simulated appliance" LED — the whole station, wired.



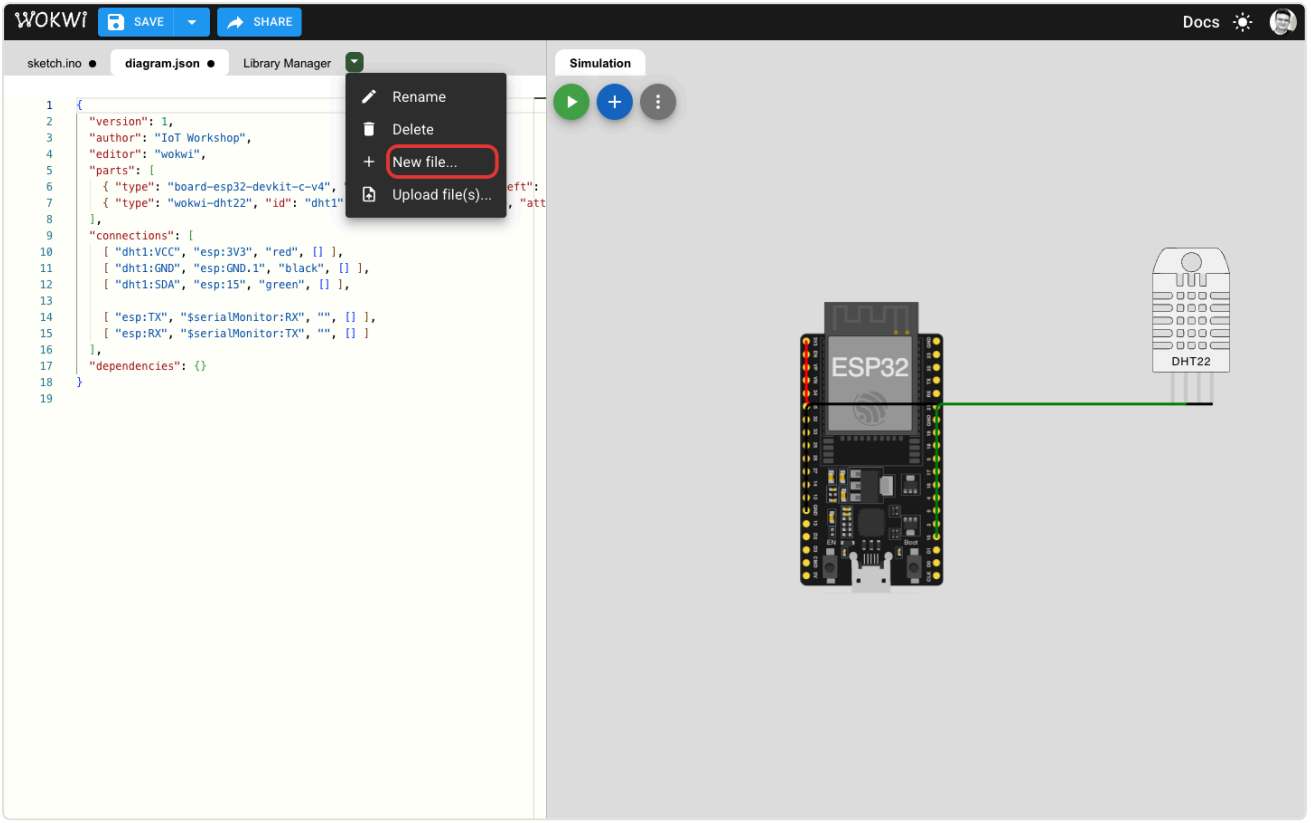
4. Add the library list

This is the heaviest lab today: four libraries.

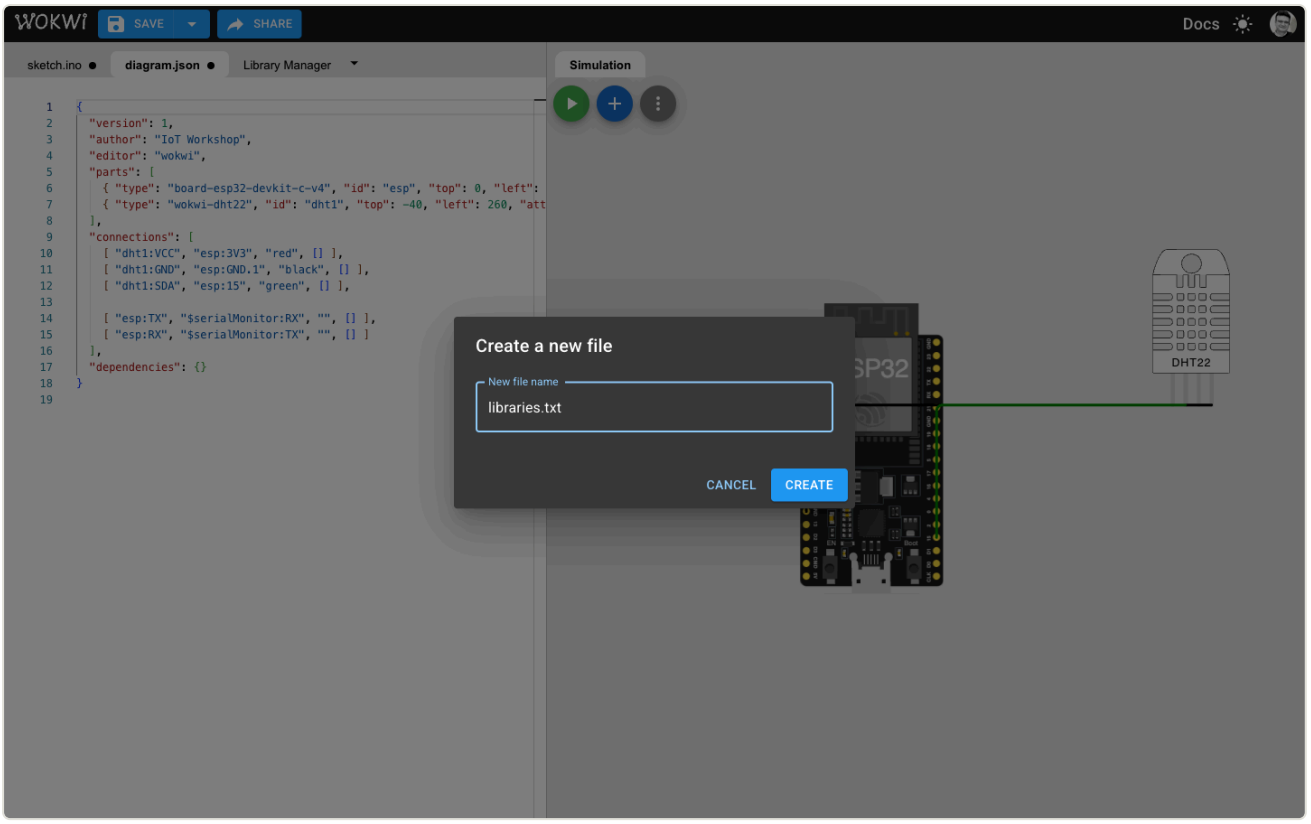
1. Click the small ▾ next to the **Library Manager** tab.



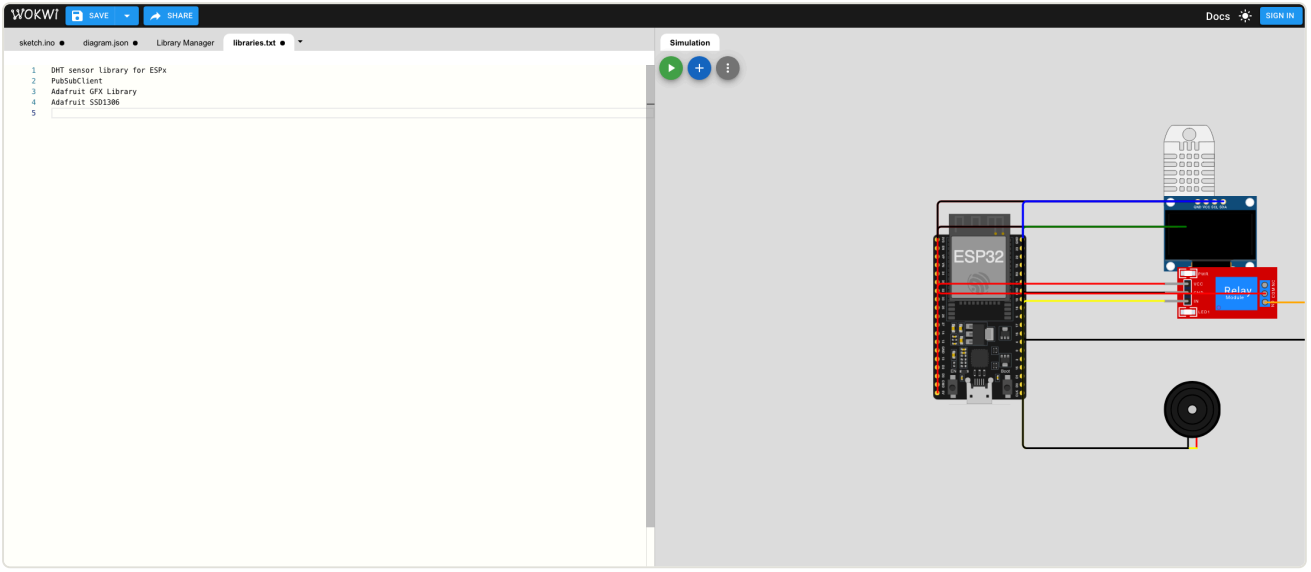
2. Choose **New file...**.



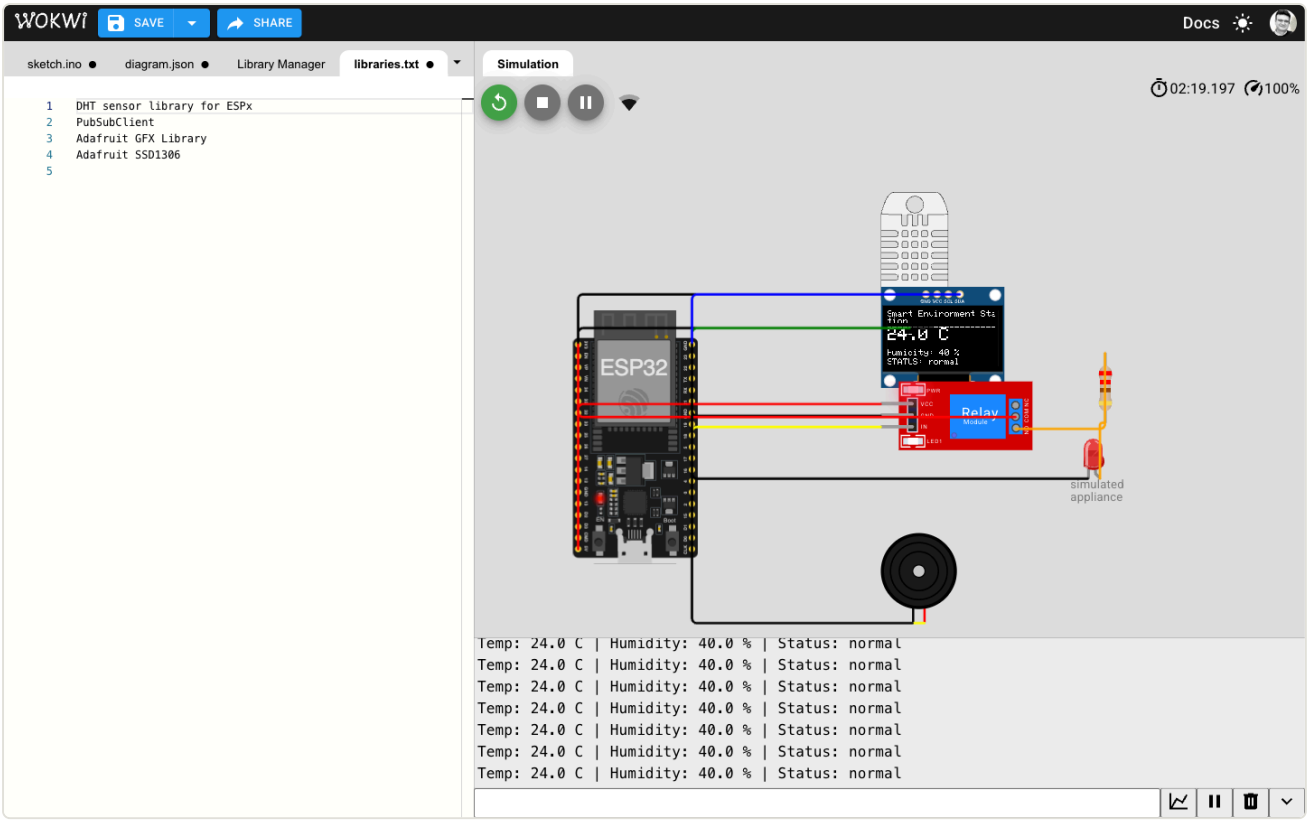
3. Type the name exactly `libraries.txt` and click **CREATE** (or press Enter).



4. Copy the contents of `libraries.txt` from this folder and paste them into the new **libraries.txt** tab.



5. Press Play



Four libraries means the first compile is the slowest of the day — it can take a couple of minutes and may show “Build Servers Busy.” That’s normal; click Play again if it happens. Once it’s running, the small OLED screen in the diagram updates with live readings and a status line.

What you should see

Serial Monitor:

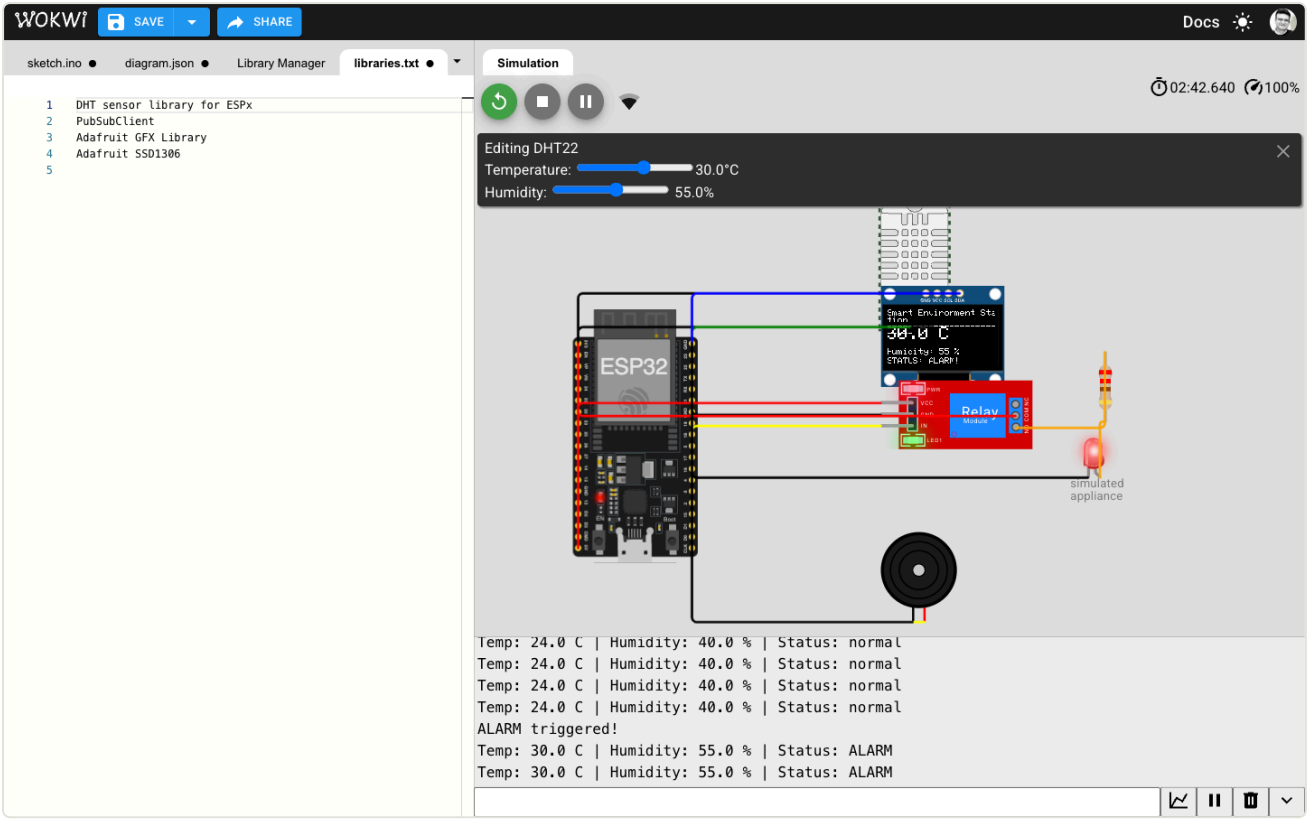
```
Connecting to Wi-Fi..... connected!
MQTT connected!
Capstone project started.
Temp: 24.0 C | Humidity: 40.0 % | Status: normal
Temp: 24.0 C | Humidity: 40.0 % | Status: normal
```

The OLED screen itself shows:

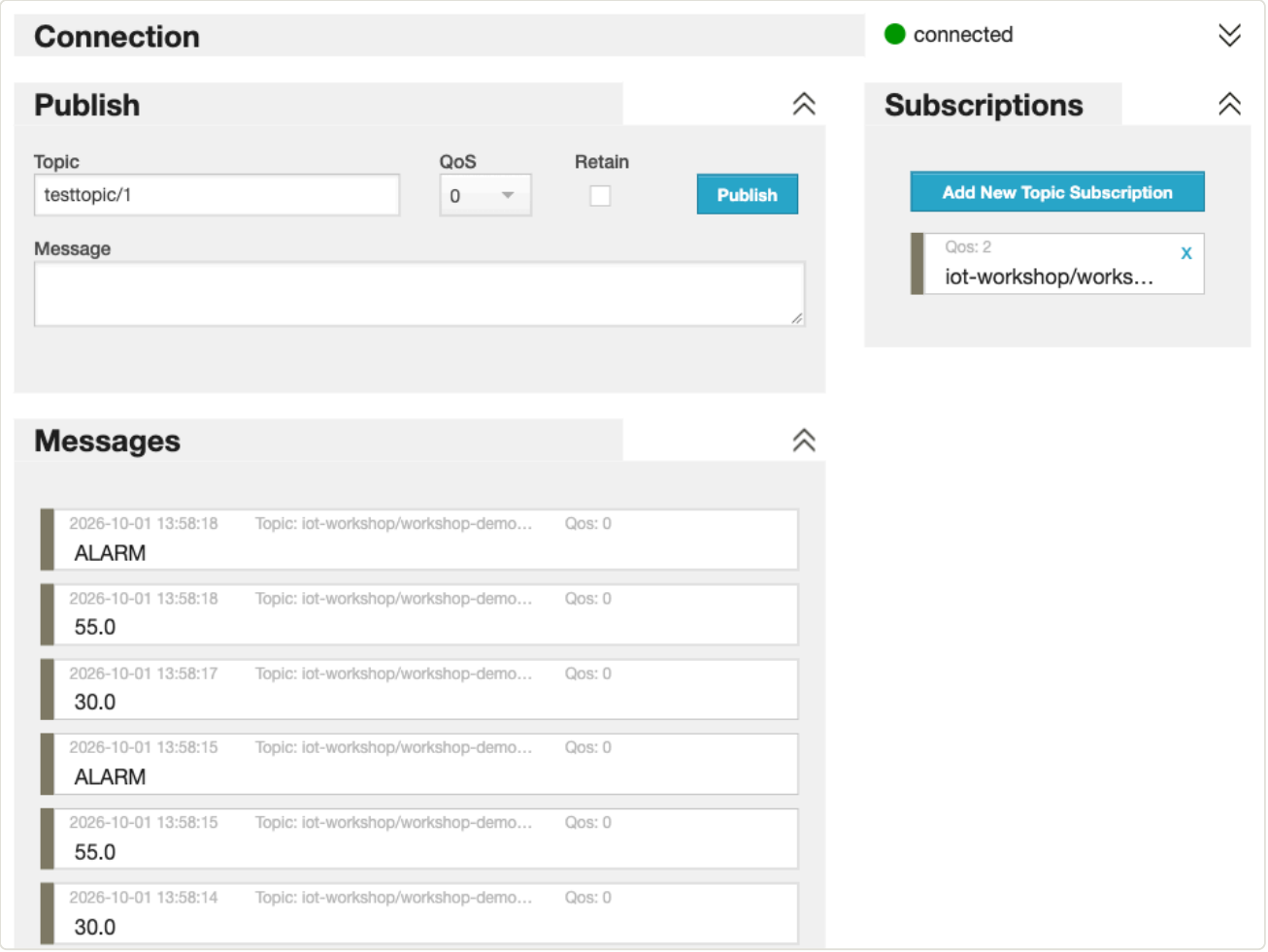
```
Smart Environment Station
24.0 C
Humidity: 40 %
STATUS: normal
```

Try it

- Click the DHT22 sensor and push the temperature above 28°C. Watch, at the same time: the OLED screen's status line change to `ALARM!`, the relay and buzzer switch on, and (if you have the HiveMQ web client open from Lab 3, subscribed to `iot-workshop/YOUR_NAME/status`) a message arrive saying `ALARM`.



In the HiveMQ web client, subscribing to `iot-workshop/YOUR_NAME/#` (the `#` means "every topic below this one") shows all three topics at once: temperature, humidity and status.



- Show your working project to the person next to you. Ask them to guess what each wire and each library is doing, before you tell them.

Troubleshooting

- **`Build failed! ... No such file or directory` for any library.** Re-check `libraries.txt` has all four lines exactly: `DHT sensor library for ESPx`, `PubSubClient`, `Adafruit GFX Library`, `Adafruit SSD1306`.
- **A blank/garbled OLED screen.** Almost always one of the four OLED wires (VCC, GND, SDA, SCL) connected to the wrong pin, or a library that failed to install — check the Serial Monitor for errors first.
- **MQTT won't connect.** Double check you changed `YOUR_NAME` — a stale client ID from a previous run can sometimes cause a slow first connection; just wait a few seconds and it will retry automatically.

Where to take this next

This capstone intentionally leaves out one big topic: **security** (Module 7). As a final exercise, discuss with your neighbor: what would you need to change before you'd trust this project to run in your own home? (Hints: the broker is public and unauthenticated; the Wi-Fi network has no

password; nothing here checks *who* is sending commands.) Then look at `docs/resources.md` for where to learn the fixes.